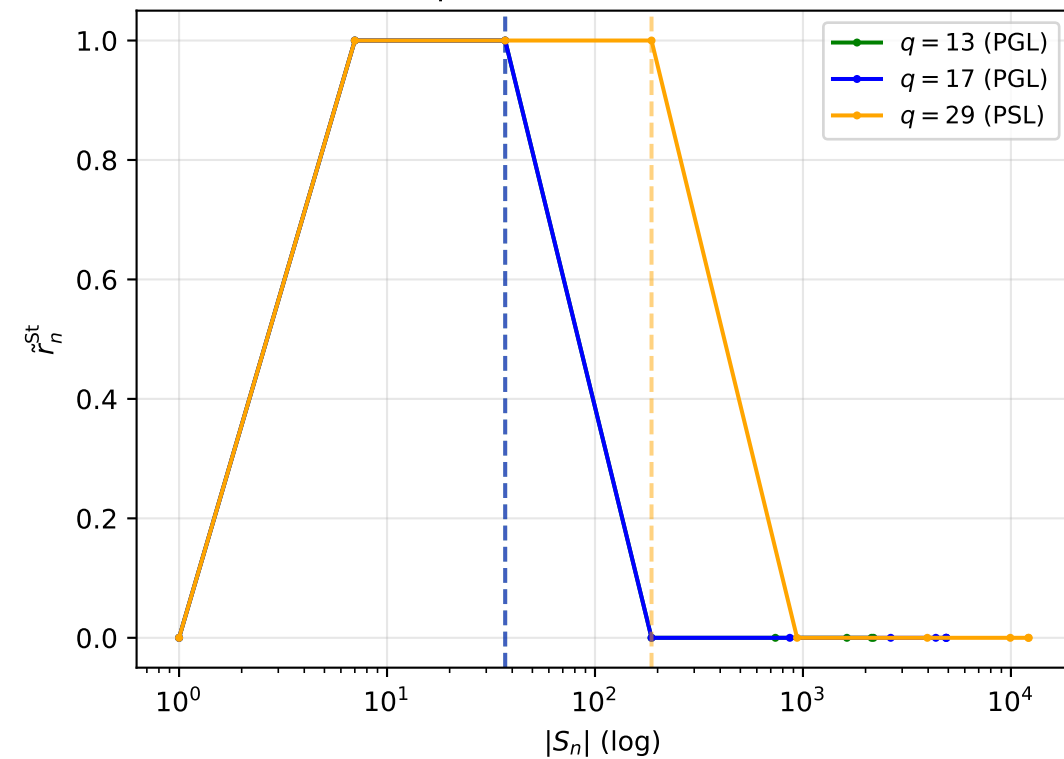
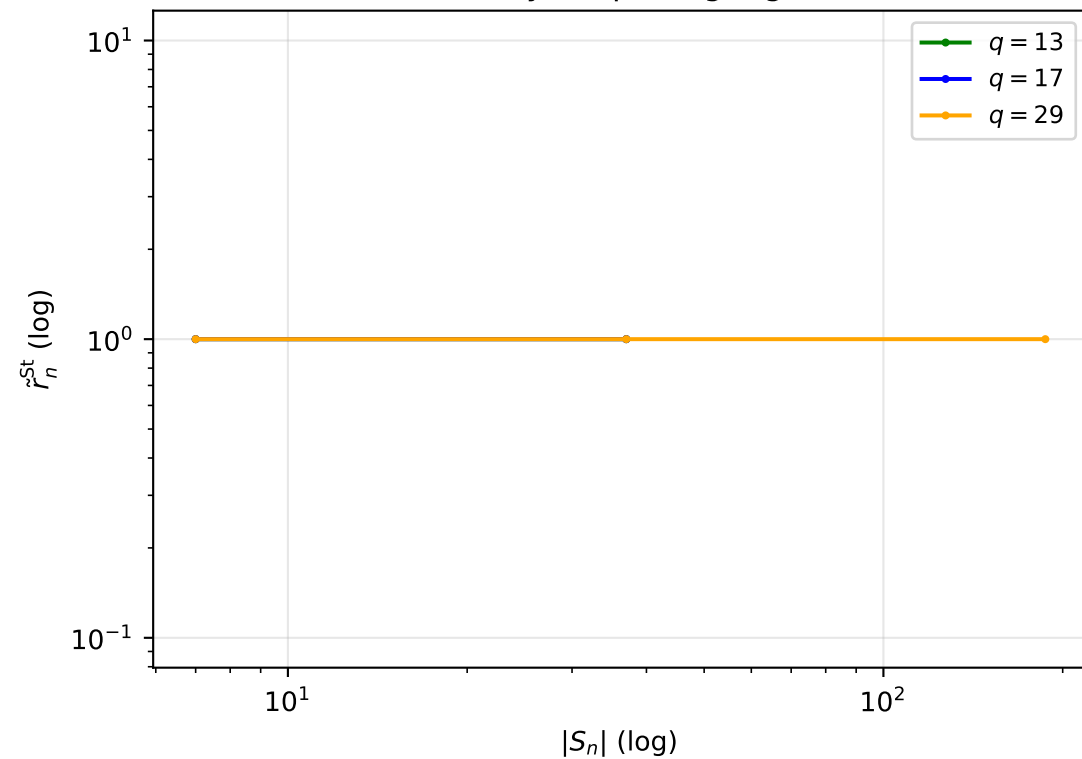


Steinberg $\tilde{\sigma}_u \odot \tilde{\sigma}_s$ pre-saturation law -- $X^{5,q}$, $q \in 13, 17, 29$, shell-layered

Trace-productive frontier fraction



Decay shape (log-log)



Cumulative trace-productive front (reference only; no exponent fit)

